

Epic 3: Data Pre-processing

Story 2: Outlier Detection and Treatment

Outlier detection and treatment are essential preprocessing steps because extreme values can significantly affect the performance and accuracy of machine learning models. Outliers may arise due to measurement errors, unusual weather conditions, or data entry mistakes, leading to biased model predictions if left untreated.

In this project, outliers were identified using **Box Plots** and the **Interquartile Range (IQR)** method. The IQR is calculated as the difference between the third quartile (Q3) and the first quartile (Q1). Values falling below the lower bound or above the upper bound are considered outliers.

Instead of removing these observations, the **IQR-based Capping** technique was applied to preserve the dataset size and maintain valuable information. Values exceeding the upper limit were replaced with the upper boundary, while values below the lower limit were replaced with the lower boundary. This approach reduced the influence of extreme values without losing important records.

Outcome

- Identified outliers using Box Plots and the IQR method.
- Applied IQR-based capping to handle extreme values.
- Preserved the original dataset size while improving data quality.
- Prepared a cleaner dataset for accurate machine learning model training.

Figure 8: Detection and treatment of outliers using Box Plots and the IQR-based capping method

```
#independent features
x=dataset.iloc[:,2:7].values

#dependent feature
y=dataset.iloc[:,9:].values
```

